# Lab 09: WLAN vs Wireless

WRT300N · Router0 (2911) · Switch0 · Printer0 · 3 PCs · Laptop0

**Lab #09** | **Topic:** Wireless Networking | **Tool:** Cisco Packet Tracer | **Cert:** CompTIA Network+

## 1. Lab Objectives

- Build a combined wired LAN and wireless WLAN in a single Packet Tracer topology.
- Configure Linksys WRT300N with SSID 'Yasin', WPA2-PSK and passphrase 'ahmed1234'.
- Connect Laptop0 wirelessly to WRT300N on the 192.168.2.0/24 subnet.
- Connect PC0, PC1, PC2 and Printer0 to Switch0 on the wired 192.168.1.0/24 subnet.
- Configure Router0 (2911) with correct static route to enable wired-to-wireless communication.

## 2. Network Topology

Switch0 (2960-24TT): Printer0 on port 0, PC0 on Fa0/1, PC1 on Fa0/2, PC2 on Fa0/3. Switch0 uplinks to Router0 (2911) on Fa0/4 → Gig0/0. Router0 Gig0/1 connects to WRT300N (port 0/0). Laptop0 connects wirelessly to the WRT300N via SSID 'Yasin'.

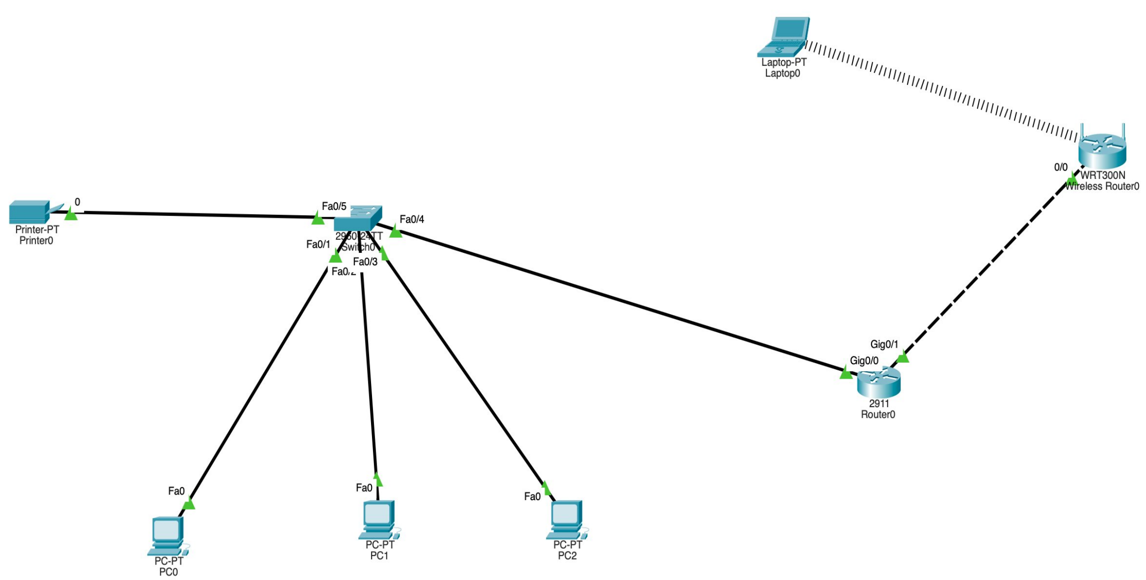


*Figure 1 – Full topology: wired LAN (Switch0, PCs, Printer) and wireless (WRT300N, Laptop0)*

## 3. IP Addressing Table

| Device | Interface | IP Address | Subnet Mask | Gateway | Network |
| --- | --- | --- | --- | --- | --- |

| | | | | | |
|---|---|---|---|---|---|
| Router0 | Gig0/0 | 192.168.1.1 | 255.255.255.0 | — | 192.168.1.0/24 |
| Router0 | Gig0/1 | 192.168.1.2 | 255.255.255.0 | — | 192.168.1.0/24 |
| WRT300N | LAN IP | 192.168.2.1 | 255.255.255.0 | 192.168.1.2 | 192.168.2.0/24 |
| PC0 | Fa0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 | 192.168.1.0/24 |
| PC1 | Fa0 | 192.168.1.20 | 255.255.255.0 | 192.168.1.1 | 192.168.1.0/24 |
| PC2 | Fa0 | 192.168.1.30 | 255.255.255.0 | 192.168.1.1 | 192.168.1.0/24 |
| Printer0 | Port 0 | 192.168.1.40 | 255.255.255.0 | 192.168.1.1 | 192.168.1.0/24 |
| Laptop0 | Wireless | DHCP (192.168.2.100) | 255.255.255.0 | 192.168.2.1 | 192.168.2.0/24 |

## 4. WRT300N Configuration

### 4.1 Basic Setup – Network & DHCP

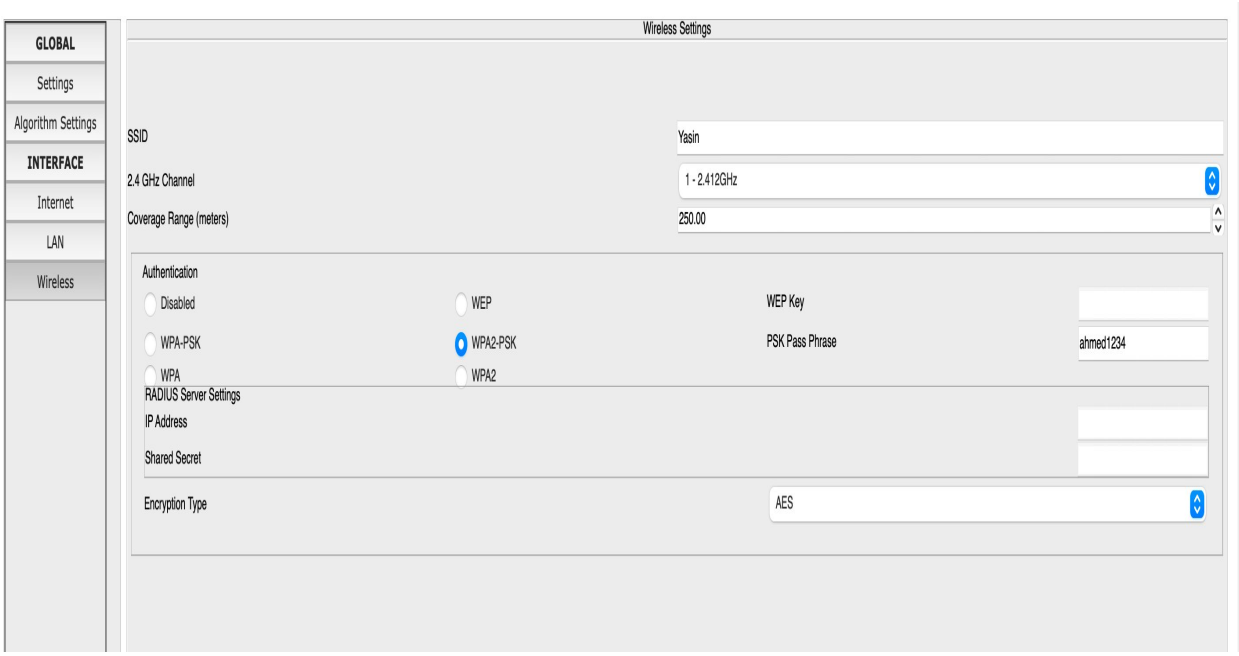


*Figure 2 – WRT300N Basic Setup: Router IP 192.168.2.1, DHCP 192.168.2.100–149*

| | |
|---|---|
| **Router IP** | 192.168.2.1 |
| **Subnet Mask** | 255.255.255.0 |
| **DHCP Server** | Enabled |
| **Start IP** | 192.168.2.100 |
| **Max Users** | 50 |
| **IP Range** | 192.168.2.100 – 192.168.2.149 |

### 4.2 Wireless Settings

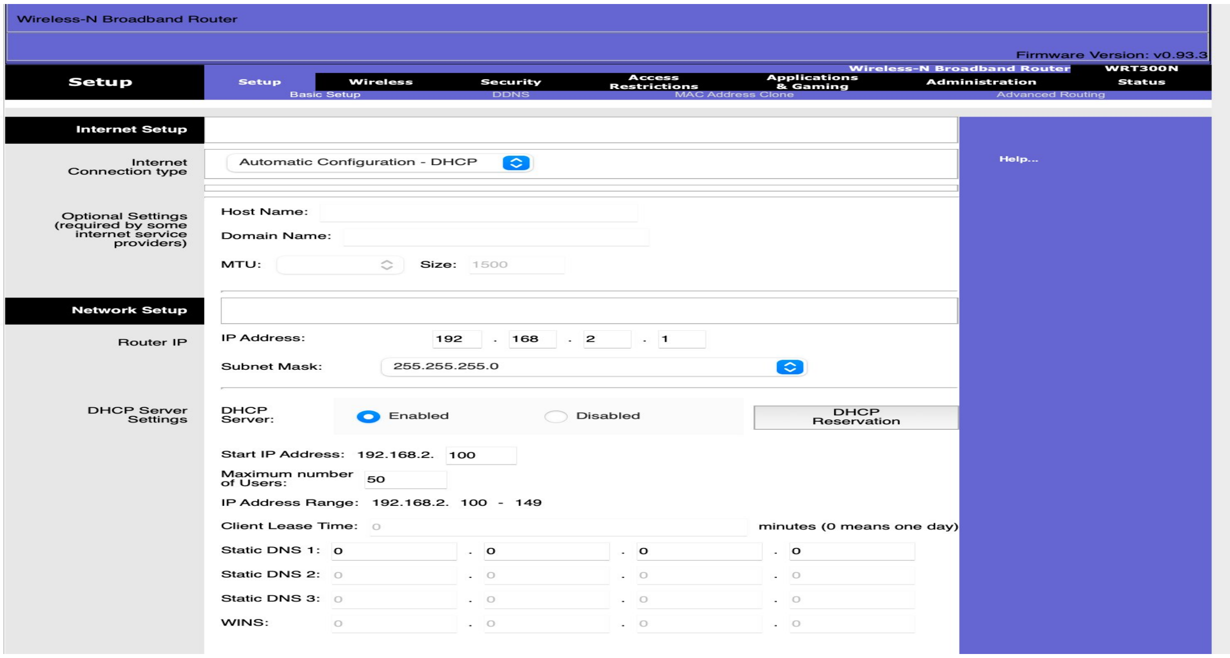


*Figure 3 – Wireless: SSID 'Yasin', Mixed mode, Channel 1 (2.412 GHz), Broadcast ON*

| | |
|---|---|
| **SSID** | Yasin |
| **Network Mode** | Mixed |
| **Standard Channel** | 1 – 2.412 GHz |
| **SSID Broadcast** | Enabled |

## 4.3 Wireless Security

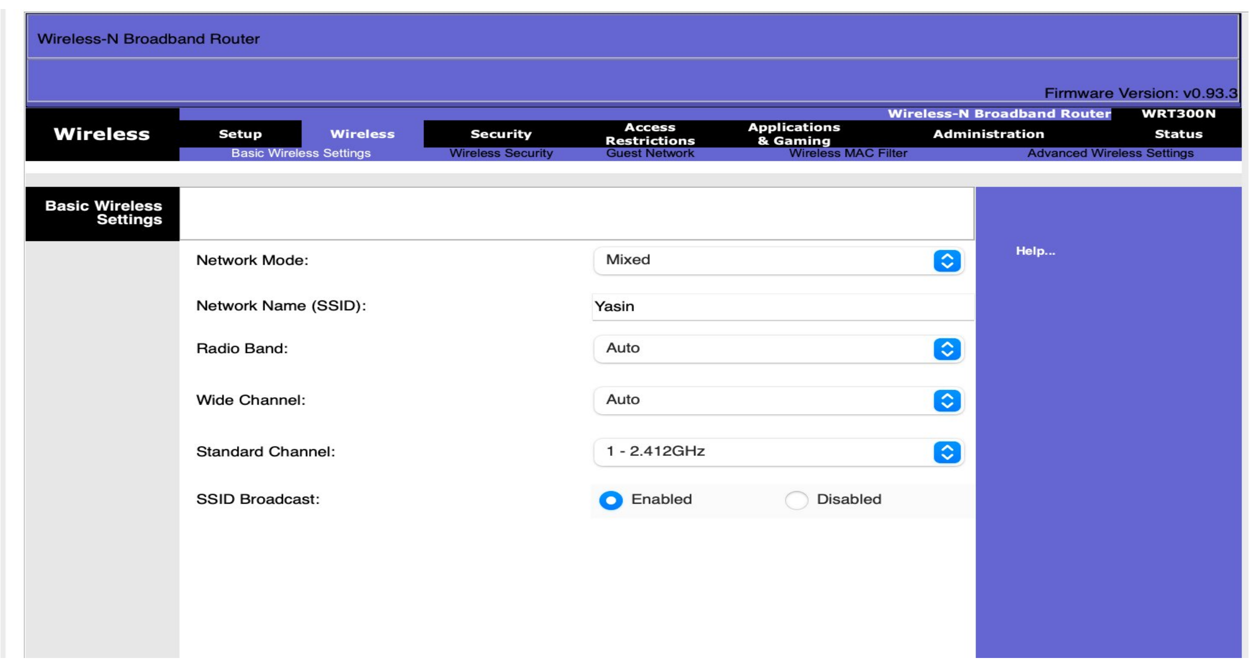


*Figure 4 – Security: WPA2 Personal, AES, Passphrase ahmed1234*

| | |
|---|---|
| **Security Mode** | WPA2 Personal |
| **Encryption** | AES |
| **Passphrase** | ahmed1234 |
| **Key Renewal** | 3600 seconds |

### 4.4 Packet Tracer Wireless Interface

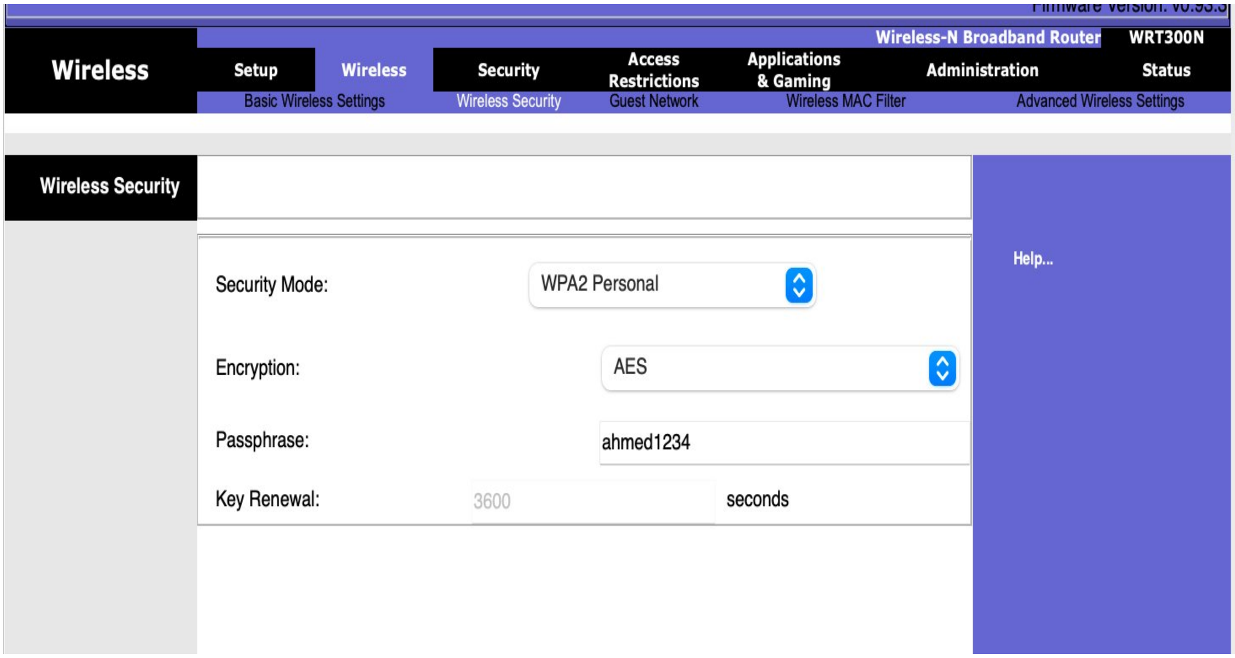


*Figure 5 – PT Wireless interface: SSID Yasin, WPA2-PSK, AES, PSK ahmed1234*

## 5. Router0 (2911) Configuration

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.1.2 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

! Static route: to reach wireless subnet 192.168.2.0/24, next hop is WRT300N WAN IP
Router0(config)#ip route 192.168.2.0 255.255.255.0 192.168.2.1
Router0(config)#exit
Router0#write memory
```

## 6. Verification

```
Laptop0>ping 192.168.2.1        ! Laptop to WRT300N gateway — expect Reply
Laptop0>ping 192.168.1.10       ! Wireless to wired PC0 (via Router0) — expect Reply
PC0>ping 192.168.2.100          ! Wired PC0 to wireless Laptop0 — expect Reply
Router0#show ip route           ! S 192.168.2.0 static route must appear
```

## 7. Key Takeaways

- The WRT300N is a home wireless router with built-in DHCP — different from an enterprise AP.
- SSID 'Yasin' is broadcast so clients can discover and join the wireless network automatically.
- WPA2 Personal + AES encryption is the correct security setting for a small wireless network.
- A static route on Router0 is needed so wired devices know how to reach the 192.168.2.0/24 wireless subnet.
- The printer on the wired LAN is accessible from wireless clients only when the static route is in place.